

Design and Development of a Modular High-Angle Steering Kit with Adjustable Ackermann Characteristics for Nissan 350Z Drift Applications

Course Report
Chassis Design

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Abstract

Drifting is a vehicle maneuver that requires large steering angles and precise control over lateral vehicle dynamics. Conventional passenger vehicle steering systems are generally designed based on standard Ackermann geometry, which may not provide the desired behavior during high-angle drift maneuvers. Therefore, adjustable steering geometry is frequently considered in drift-oriented vehicle development.

In this project, a modular high-angle steering kit was designed and developed for a Nissan 350Z platform to provide adjustable Ackermann steering characteristics. The proposed system enables modification of the steering linkage geometry to achieve different Ackermann characteristics, including configurations approaching parallel steering and reverse Ackermann behavior by utilizing interchangeable steering inserts.

The development process included reverse engineering of the original steering knuckle using three-dimensional scanning and Geomagic software, followed by CAD reconstruction and mechanical design of the modified steering components. The steering system was redesigned to increase the maximum steering angle while preventing mechanical interference between steering components at large steering angles. The final design achieved a maximum steering angle of 41 degrees.

The designed components were manufactured using CNC machining, laser cutting, and CNC bending processes. The performance of different steering geometries was evaluated using CarSim simulations under identical driving conditions. Vehicle trajectory, steering angle, yaw rate, sideslip angle, vehicle sideslip angle (β), and steering torque were analyzed to investigate the influence of Ackermann geometry variation on drift vehicle behavior.

1. Introduction

Drifting is a vehicle maneuver in which the driver intentionally maintains the vehicle in a controlled oversteering condition while generating a significant lateral slip angle. Unlike conventional driving conditions, drift maneuvers require precise control of lateral vehicle dynamics, steering response, and tire behavior. Therefore, modifications to standard vehicle systems are commonly performed to improve controllability and stability during high-angle cornering.

The steering system plays a critical role in drift vehicle performance. Production vehicles are generally designed based on conventional Ackermann steering geometry, which provides suitable tire alignment during normal cornering conditions. However, during drifting, the vehicle operates

at significantly higher steering angles and larger tire slip angles, where conventional steering geometry may not provide the desired handling characteristics.

Ackermann steering geometry defines the relationship between the steering angles of the inner and outer wheels during cornering. By modifying the steering arm geometry, different steering characteristics such as parallel steering and reverse Ackermann can be achieved. These configurations can influence the vehicle response during drift maneuvers by changing the relationship between tire steering angles and vehicle motion.

The objective of this project was to design and develop a modular high-angle steering kit for a Nissan 350Z drift application. The proposed design aimed to increase the maximum steering angle and provide adjustable Ackermann characteristics through interchangeable steering inserts. The development process included reverse engineering of the original steering knuckle, CAD-based mechanical design, prototype manufacturing, and vehicle dynamics evaluation using CarSim simulations.

2. Steering Geometry Fundamentals

2.1. Steering Geometry Overview

The steering system is one of the most important subsystems affecting vehicle handling and directional control. During a cornering maneuver, the front wheels of a vehicle do not follow the same trajectory. Since the inner wheel travels along a smaller radius compared with the outer wheel, the steering system must provide different steering angles for the two wheels to minimize tire lateral slip.

The geometric relationship between the steering components determines the steering characteristics of a vehicle. Important parameters such as the position of the steering arm, kingpin axis, and tie-rod attachment point directly influence the relationship between the inner and outer wheel steering angles. Therefore, modifying these parameters can significantly affect vehicle behavior, especially under high steering angles and large tire slip conditions.

In conventional passenger vehicles, the steering geometry is primarily designed for normal driving conditions, where maintaining tire alignment and reducing tire wear are the main objectives. However, drift vehicles operate under different conditions, where large steering angles, high lateral slip angles, and rapid changes in vehicle yaw motion require different steering characteristics

2.2. Ackermann Steering Geometry

Ackermann steering geometry is a widely used steering configuration designed to provide different steering angles for the inner and outer front wheels during cornering. In an ideal Ackermann condition, the extensions of the steering arms intersect at the center of the rear axle. This geometric relationship ensures that all wheels approximately follow concentric circular paths during a turn.

In this configuration, the inner wheel rotates to a larger steering angle than the outer wheel because it travels along a smaller turning radius. This behavior reduces tire scrub during low-speed cornering and improves vehicle efficiency under normal driving conditions.

Figure 1 illustrates the ideal Ackermann steering geometry, where the steering arm extensions intersect at the rear axle center.

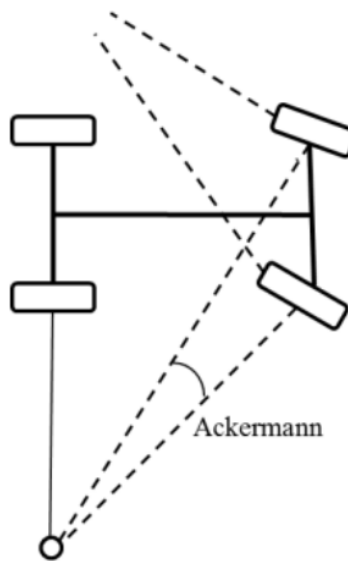


Figure 1 Ideal Ackermann steering geometry.

Although Ackermann geometry provides desirable characteristics for conventional driving, its behavior may not be optimal for drifting applications. During drifting, the vehicle operates with significant tire slip angles, and the relationship between steering input and vehicle response becomes different from normal cornering conditions.

2.3. Parallel Steering Geometry

Parallel steering is a configuration in which the steering arms are arranged approximately parallel to each other. In this condition, the inner and outer wheels achieve similar steering angles during cornering.

Compared with conventional Ackermann geometry, parallel steering reduces the difference between the steering angles of the two front wheels. This characteristic can be beneficial in conditions where both tires operate with significant slip angles, such as drifting.

Since drift vehicles are controlled primarily through tire slip rather than pure geometric rolling, parallel steering can provide different handling characteristics compared with standard Ackermann steering.

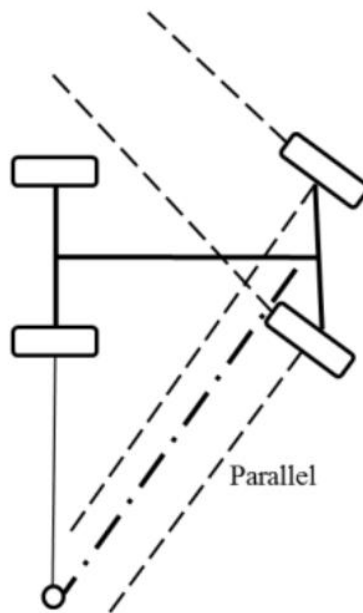


Figure 2 Parallel steering geometry

2.4. Reverse Ackermann Steering Geometry

Reverse Ackermann geometry is an alternative steering configuration in which the extensions of the steering arms intersect at a point located in front of the vehicle rather than at the rear axle center.

This arrangement results in a smaller steering angle difference between the inner and outer wheels compared with conventional Ackermann geometry. Under high-slip-angle conditions, such as those experienced during drifting, reverse Ackermann characteristics can provide different tire behavior and vehicle response.

The suitability of reverse Ackermann geometry depends on several factors, including vehicle configuration, tire characteristics, suspension design, and driving conditions. Therefore, evaluating different steering geometries through vehicle dynamics simulation is necessary to understand their effects on drift performance.

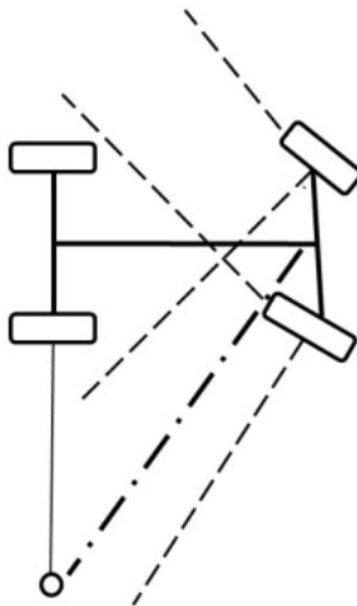


Figure 3 Reverse Ackermann steering geometry

2.5. Steering Geometry Considerations for Drift Applications

Unlike normal cornering, drifting involves maintaining a controlled lateral slip condition where the vehicle is intentionally operated beyond the limits of conventional tire behavior. Therefore, steering system modifications for drift applications usually focus on increasing steering angle capability and adjusting the relationship between steering input and vehicle response.

High-angle steering systems commonly require modifications to the steering linkage to avoid mechanical interference between components at large steering angles. Additionally, adjustable

steering geometries allow the vehicle setup to be adapted for different tracks, driving styles, and vehicle behaviors.

In this project, the steering geometry was modified through an adjustable linkage mechanism that changes the effective position of the tie-rod attachment point relative to the steering axis. This approach allows different Ackermann characteristics to be investigated and provides a flexible solution for drift-oriented steering development.

3. Project Objectives

The main objective of this project was to design and develop a modular high-angle steering kit for a Nissan 350Z drift application. The project focused on improving the steering capability of the vehicle while providing the possibility of modifying steering geometry characteristics for different drift conditions.

The standard steering system of production vehicles is generally designed for conventional driving conditions and may not provide sufficient steering angle capability or suitable steering characteristics for drift applications. Therefore, a modified steering system was developed to address the requirements of high-angle drifting.

The main objectives of this project were defined as follows:

1. Increasing Maximum Steering Angle

The first objective was to increase the maximum steering angle of the front wheels by modifying the steering linkage geometry. The designed system aimed to achieve larger steering angles while maintaining compatibility with the original steering knuckle and suspension components.

2. Preventing Mechanical Interference at High Steering Angles

Increasing steering angle can lead to interference between steering components, particularly between the tie rod, steering knuckle, and suspension elements. Therefore, the steering linkage was redesigned to provide sufficient clearance during high-angle steering conditions.

3. Developing an Adjustable Steering Geometry Mechanism

A modular adjustment mechanism was developed to modify the effective tie-rod attachment position relative to the steering axis. This design enables variation of steering characteristics by changing the steering linkage geometry without replacing the complete steering assembly.

4. Manufacturing and Testing the Designed Prototype

The designed components were manufactured using appropriate fabrication methods, including CNC machining, laser cutting, and CNC bending processes. The manufactured prototype was assembled and tested to verify the functionality of the developed steering system.

5. Evaluating Different Ackermann Characteristics Using Vehicle Simulation

The influence of different steering geometries on vehicle behavior was investigated using CarSim simulations. Several steering configurations, including conventional Ackermann, parallel steering, and reverse Ackermann geometries, were evaluated under identical driving conditions.

The simulation results were analyzed based on vehicle trajectory, steering angle, yaw rate, sideslip angle, vehicle lateral behavior, and steering torque to investigate the effect of steering geometry variation on drift vehicle dynamics.

4. Reverse Engineering Process

4.1. 3D Scanning of the Original Steering Knuckle

In order to develop a modified steering system while maintaining compatibility with the original vehicle suspension components, the geometry of the original steering knuckle was captured through a three-dimensional scanning process.

The complete steering knuckle was scanned to obtain the geometric information required for reconstruction and redesign. The obtained scan data provided an accurate representation of the original component, including its mounting features, connection locations, and overall geometry.

The reverse engineering approach was selected to preserve the original interfaces between the steering knuckle and other suspension components while allowing modifications to the steering linkage geometry.

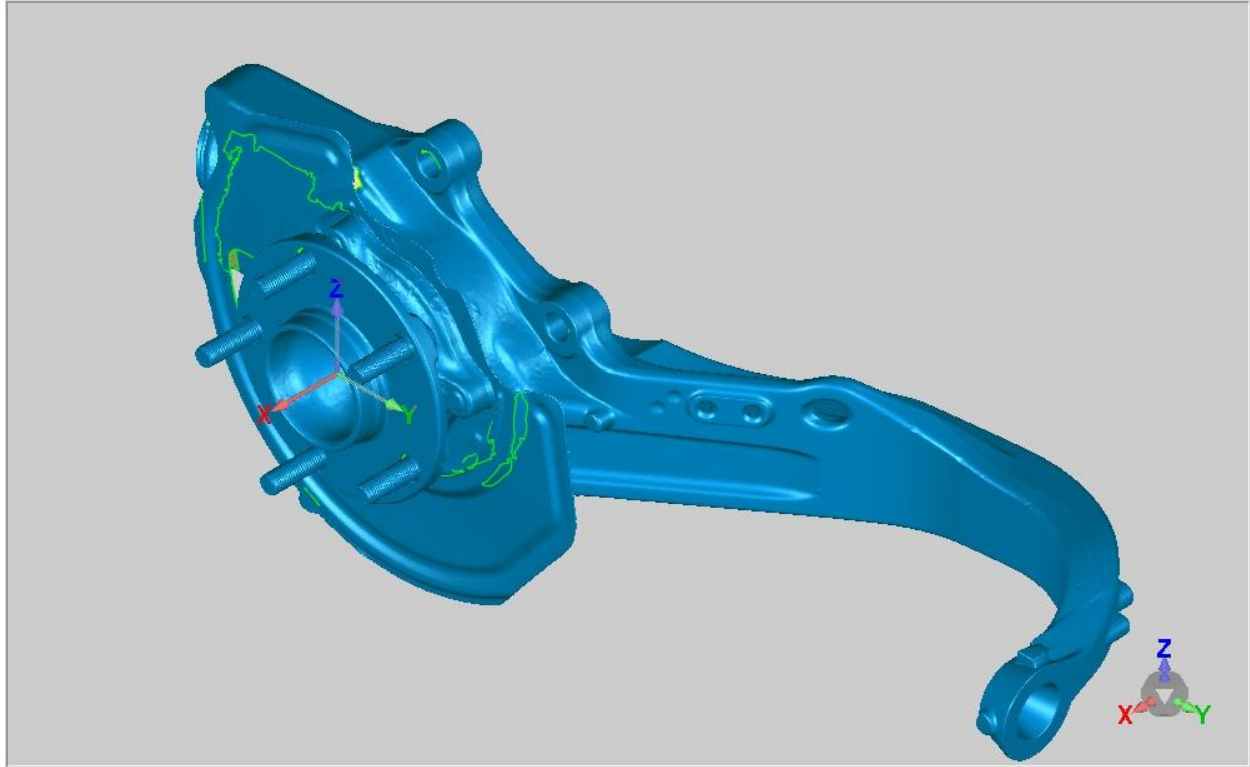


Figure 4 Three-dimensional scanning of the original steering knuckle

4.2. Point Cloud Processing Using Geomagic

The acquired scan data was imported into Geomagic for processing and preparation. The point cloud was cleaned and processed to generate a suitable reference model for CAD development.

The main objective of this stage was to extract the important geometric features of the original steering knuckle, particularly the mounting locations and surfaces required for the design of the modified steering mechanism.

The processed point cloud was then used as a reference for developing the CAD model while maintaining the original component interfaces.



Figure 5 Processed point cloud of the steering knuckle in Geomagic

4.3. CAD Reconstruction and Design Reference

After processing the scan data, the point cloud was imported into SolidWorks as the reference geometry for CAD modeling. The new steering components were designed directly based on the reconstructed geometry to ensure proper fitment with the original steering knuckle.

Using the scanned geometry as the design reference allowed the development of the steering kit without altering the original suspension mounting points. This approach reduced design uncertainties and improved the accuracy of the final prototype.

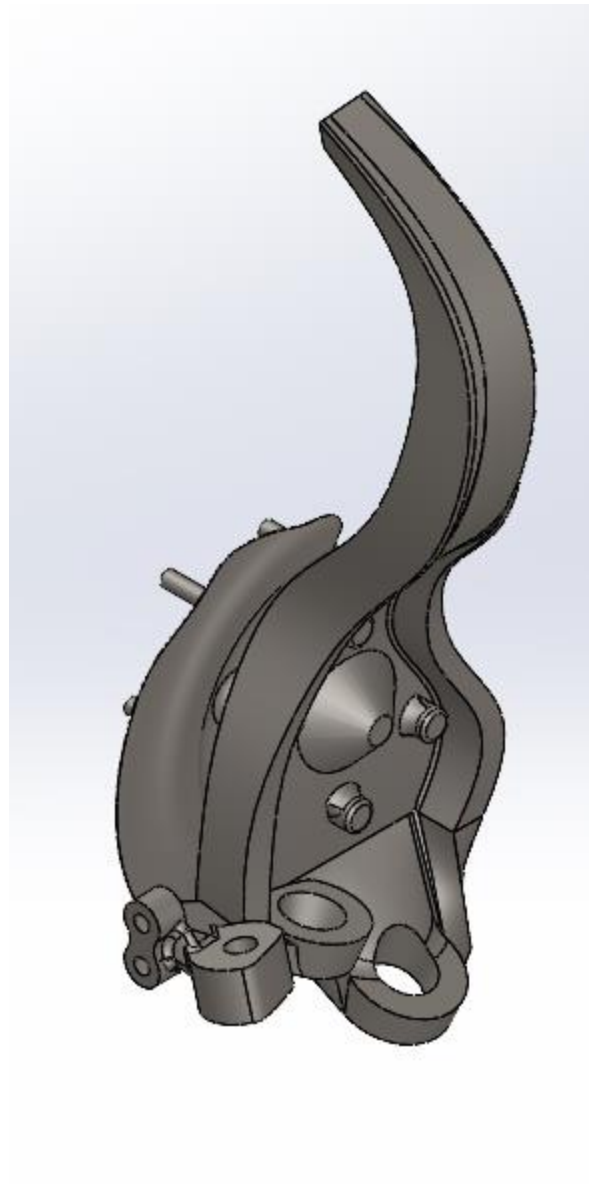


Figure 6 CAD modeling based on the reconstructed steering knuckle geometry

5. Steering Kit Design

5.1. Design Requirements

The design requirements of the proposed steering kit were defined based on the operational conditions of a drift vehicle and the limitations of the original steering system.

During drift maneuvers, the vehicle requires significantly larger steering angles compared with conventional driving conditions. However, increasing the steering angle using the original steering configuration can lead to mechanical interference between the steering components, particularly between the tie rod and the steering knuckle.

Therefore, the main design requirements of the developed steering kit were defined as follows:

- Increasing the maximum steering angle capability of the front wheels.
- Maintaining compatibility with the original Nissan 350Z steering knuckle.
- Preventing mechanical interference during high steering angles.
- Providing the capability to modify steering geometry characteristics.
- Developing a manufacturable and reliable mechanical solution.

The final design was developed based on the geometry obtained from the reverse-engineered steering knuckle, ensuring that the modified components could be integrated with the original suspension system.

5.2. Steering Geometry Modification

The steering characteristics of a vehicle are strongly affected by the relative position of the tie-rod attachment point with respect to the steering axis. Changing this position modifies the effective steering arm geometry and consequently changes the relationship between the steering angles of the inner and outer wheels.

In the original steering system, the tie-rod attachment location is fixed, resulting in a fixed steering geometry. To provide adjustable steering characteristics, a modified linkage mechanism was designed in which the tie-rod connection point could be repositioned.

The developed mechanism changes the location of the tie-rod attachment point relative to the kingpin axis, allowing different steering geometries to be generated.

5.3. Adjustable Tie-Rod Attachment Mechanism

A dedicated adjustable mechanism was developed to control the position of the tie-rod connection point. The mechanism consists of a rotating insert that is mounted on the steering assembly, where the tie-rod attachment point is located within an obround slot.

By rotating the insert, the position of the tie-rod connection point changes along the predefined path. This modification changes the effective steering arm geometry and allows adjustment of the Ackermann characteristics without replacing the complete steering assembly.

The use of an obround slot provides a continuous adjustment capability while maintaining a compact mechanical design. This approach enables different steering configurations to be investigated using the same hardware.

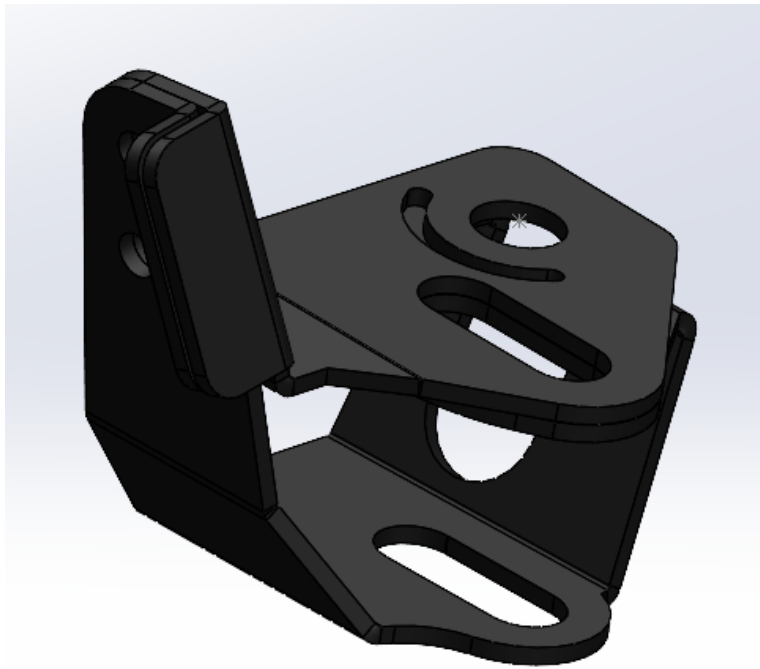


Figure 7 Adjustable tie-rod attachment mechanism and obround slot design

5.4. Effect of Tie-Rod Position on Ackermann Characteristics

The Ackermann characteristics of the steering system depend on the relationship between the steering arm orientations and the position of the tie-rod attachment points.

In conventional Ackermann steering geometry, the extensions of the steering arms intersect at the center of the rear axle. For parallel steering, the steering arms are approximately parallel, while in reverse Ackermann geometry, their extensions intersect in front of the vehicle.

In this project, these different steering characteristics were achieved by modifying the effective position of the tie-rod attachment point through the developed adjustment mechanism. The mechanism was designed based on geometric construction of the steering linkage rather than a predefined Ackermann percentage.

This approach provides flexibility for investigating different steering configurations and adapting the vehicle setup for different drift conditions.

5.5. High-Angle Steering Capability

One of the main objectives of the developed steering kit was increasing the maximum steering angle of the front wheels.

The original steering geometry was modified to provide additional clearance between steering components during large steering inputs. The final design allowed the front wheels to achieve a maximum steering angle of approximately 41 degrees.

The increased steering capability enables the driver to maintain vehicle control during high-angle drift maneuvers while reducing the risk of mechanical interference within the steering system.

5.6. Final CAD Assembly

After completing the mechanical design, the modified steering components were assembled in SolidWorks using the reverse-engineered steering knuckle as the reference model.

The final CAD assembly was evaluated to verify component compatibility, clearance, and assembly feasibility before manufacturing.

The completed design consisted of the adjustable steering mechanism, modified linkage components, and required mounting elements while preserving the original steering knuckle interfaces.

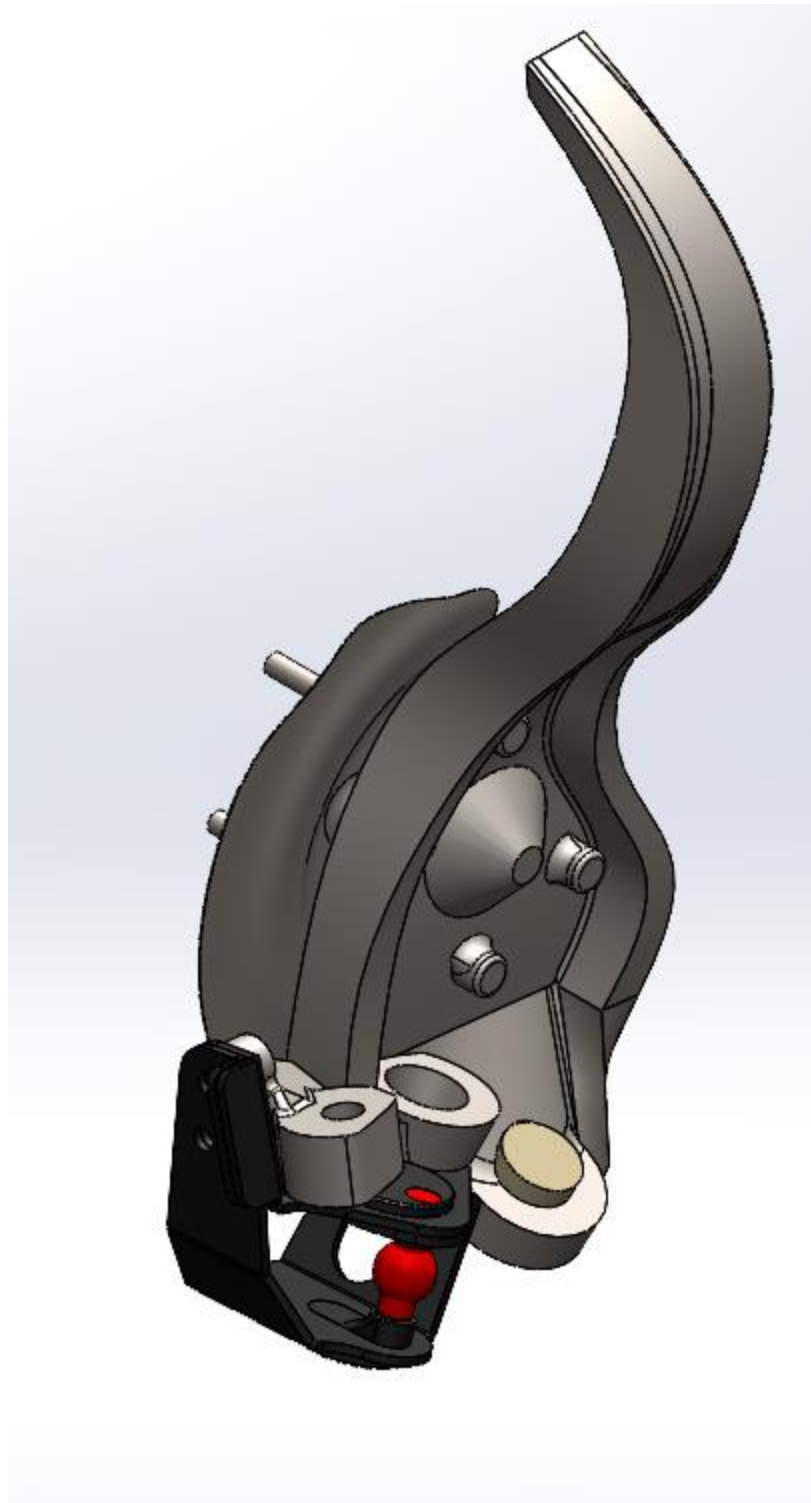


Figure 8 Final CAD model of the developed steering kit

6. Manufacturing

6.1. Manufacturing Approach

After completing the design and verification of the steering kit in the CAD environment, the designed components were manufactured to evaluate the feasibility of the proposed mechanical solution.

The manufacturing process was selected based on the geometry, functional requirements, and required dimensional accuracy of each component. Since the developed steering mechanism directly affects vehicle handling and must withstand operational loads, accurate fabrication and proper assembly of the components were considered essential.

The main manufacturing processes used in this project included CNC machining, laser cutting, and CNC bending.



Figure 9 Manufactured components of the developed steering kit

6.2. CNC Machining of Steering Components

The components responsible for controlling the steering geometry required precise machining due to their direct influence on the tie-rod position and steering characteristics.

CNC machining was used for manufacturing the adjustable steering components to achieve the required dimensional accuracy and maintain the designed geometric relationships. The machined parts were produced according to the CAD models developed based on the reverse-engineered steering knuckle geometry.

The accuracy of these components was particularly important because small variations in the tie-rod attachment location can significantly affect the resulting steering geometry.

6.3. Laser Cutting and CNC Bending

Some structural components of the steering kit were manufactured using sheet metal fabrication methods. Laser cutting was used to produce the required profiles with accurate dimensions, while CNC bending was applied to achieve the desired final geometry.

These manufacturing methods provided a suitable balance between fabrication accuracy, structural integrity, and production simplicity.

6.4. Prototype Assembly

After manufacturing, the components were assembled with the original steering knuckle to evaluate the compatibility of the designed system.

The prototype assembly process verified the integration between the modified steering mechanism and the original vehicle components. The assembled system was inspected to ensure proper movement of the steering linkage and sufficient clearance during high steering angles.



Figure 10 Final assembled prototype of the steering kit

6.5. Manufacturing Considerations

The manufacturing process demonstrated the feasibility of producing the proposed adjustable steering mechanism using commonly available fabrication methods.

The combination of CNC machining, laser cutting, and CNC bending allowed the prototype to maintain the required geometric accuracy while remaining practical for small-scale production and vehicle development applications.

7. Vehicle Dynamics Simulation

7.1. CarSim Model Development

To evaluate the influence of different steering geometries on vehicle behavior, vehicle dynamics simulations were performed using CarSim software.

A vehicle model based on the specifications of the Nissan 350Z was developed by introducing the required vehicle parameters into the simulation environment. The same vehicle configuration was maintained for all simulation cases to ensure that the observed differences were mainly caused by changes in steering geometry.

The vehicle was simulated under drift-oriented operating conditions. The vehicle was considered to be in gear with a locked differential, and full throttle input was applied during the maneuver to generate a controlled oversteering condition.

7.2. Simulation Conditions

A circular driving path with a radius of 10 meters was selected for the simulations. The relatively small corner radius was chosen to create a high steering demand condition and to investigate the behavior of different Ackermann characteristics under large steering angles.

For all simulation cases, the following parameters were kept constant:

- Vehicle model and physical parameters
- Driving path
- Throttle input
- Differential condition
- Initial vehicle conditions

Only the front wheel steering angle relationship was modified to represent different steering geometries.

Table 1 Simulation conditions

<i>Parameter</i>	<i>Value</i>
<i>Vehicle</i>	Nissan 350Z (vehicle parameters)
<i>Simulation Software</i>	CarSim
<i>Driving Path</i>	Circular path
<i>Path Radius</i>	10 m
<i>Vehicle Condition</i>	2 nd gear
<i>Differential</i>	Locked differential
<i>Throttle Input</i>	Full throttle (100%)
<i>Steering Configurations</i>	Conventional Ackermann, Parallel Steering, Reverse Ackermann
<i>Steering Geometry Implementation</i>	Different inner and outer wheel steering angles
<i>Constant Parameters</i>	Vehicle parameters, driving path, throttle input, differential condition
<i>Evaluated Outputs</i>	Vehicle trajectory (X-Y), vehicle sideslip angle (β), tire slip angle, steering angle, steering torque, yaw rate

7.3. Implementation of Steering Geometries

Since the objective of the simulation was to investigate the influence of Ackermann characteristics on vehicle behavior, different steering configurations were implemented by defining different relationships between the inner and outer front wheel steering angles.

Three steering cases were evaluated:

1. Conventional Ackermann steering
2. Parallel steering
3. Reverse Ackermann steering

The required wheel steering angles for each configuration were calculated based on steering geometry relationships and then applied in the CarSim model.

This approach allowed the effect of different steering characteristics to be compared while maintaining identical vehicle and driving conditions.

7.4. Simulation Outputs

The effect of steering geometry variation was evaluated using several vehicle dynamic outputs obtained from CarSim.

The following parameters were analyzed:

Vehicle Trajectory (Y-X Path)

The vehicle trajectory was evaluated to compare the ability of each steering configuration to follow the predefined circular path.

Vehicle Sideslip Angle (β)

The vehicle sideslip angle was analyzed as an important indicator of drift behavior and lateral vehicle motion.

Tire Slip Angle

The tire slip angle was investigated to evaluate the effect of steering geometry on tire operating conditions during cornering.

Steering Angle

The steering input required for maintaining the drift condition was compared between different steering configurations.

Steering Torque

The steering torque was analyzed to investigate the effect of steering geometry changes on driver steering effort.

Yaw Rate

The yaw rate response was evaluated to compare the rotational behavior of the vehicle under different steering configurations.

7.5. Simulation Methodology Summary

The simulation methodology provided a controlled environment for comparing different steering geometries. By maintaining identical vehicle conditions and modifying only the front wheel steering angle relationship, the influence of Ackermann characteristics on drift vehicle behavior could be investigated.

8. Results and Discussion

8.1. Vehicle Trajectory Analysis

The vehicle trajectory obtained from CarSim simulations was analyzed to compare the ability of different steering geometries to maintain the desired drift path.

Under identical driving conditions, including the same circular path, throttle input, and vehicle configuration, the steering geometries exhibited different path-following behaviors.

The reverse Ackermann configuration maintained the vehicle within the desired trajectory for the longest duration compared with the other configurations. The ideal Ackermann configuration showed the second-best path stability, while the parallel steering configuration resulted in the least capability to maintain the predefined path.

The results indicate that under the investigated high-slip-angle drift condition, reducing the difference between inner and outer wheel steering angles through reverse Ackermann characteristics can contribute to improved vehicle path control.

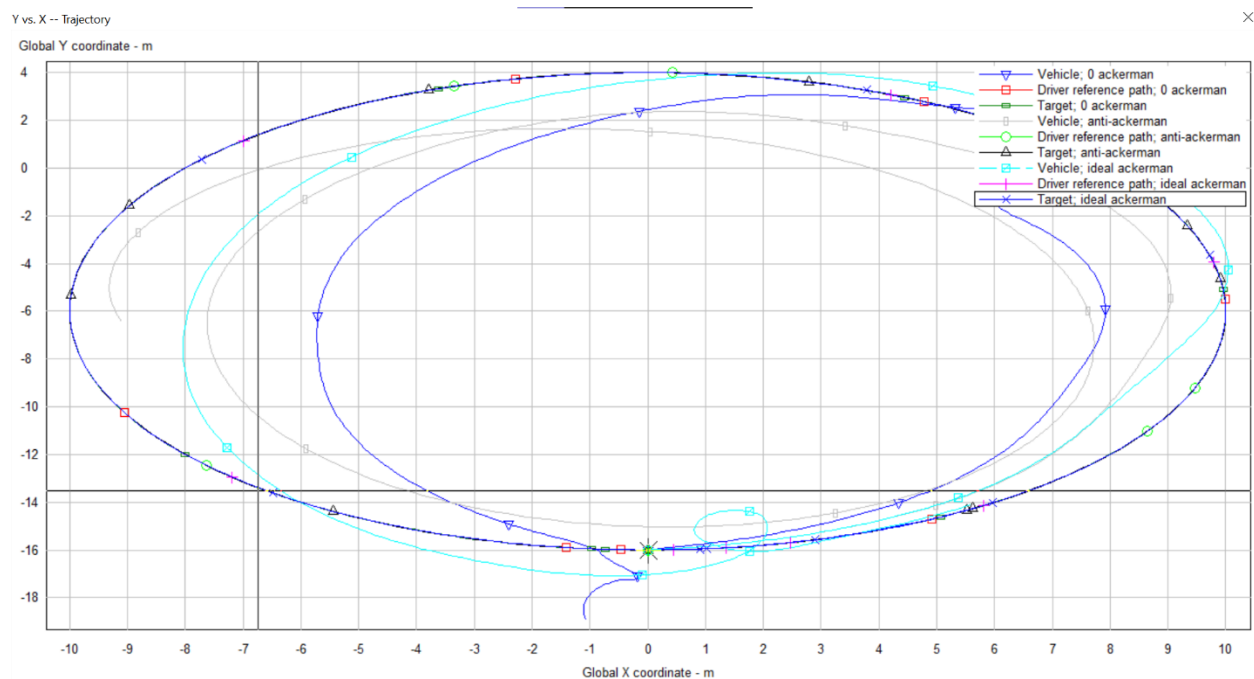


Figure 11 Vehicle trajectory comparison for different steering geometries

8.2. Vehicle Sideslip Angle Analysis

The vehicle sideslip angle (β) was analyzed as an important parameter for evaluating drift stability.

The reverse Ackermann configuration demonstrated the smallest variation in sideslip angle compared with the other steering geometries. A lower variation in β indicates a more consistent lateral motion state and a more stable drift condition.

In contrast, the parallel steering configuration showed larger fluctuations in sideslip angle, indicating less stable vehicle behavior during the simulated maneuver.

These results suggest that the reverse Ackermann configuration provided a more controlled lateral response under the selected drift conditions.

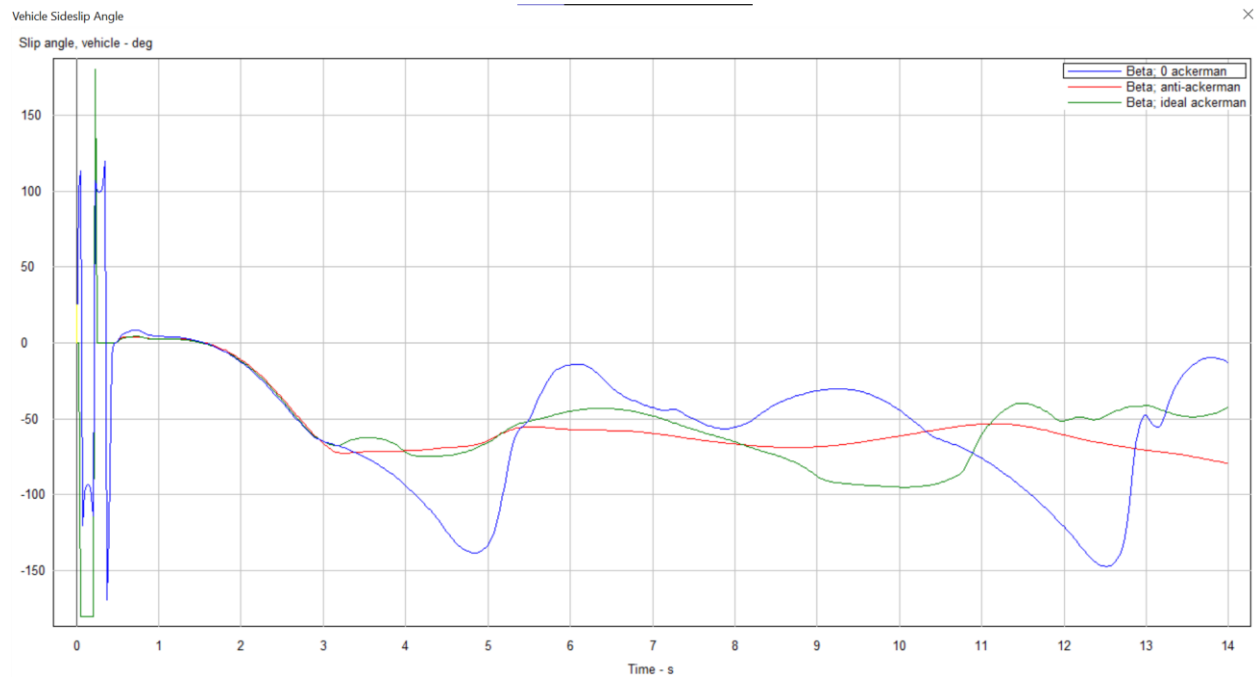


Figure 12 Vehicle sideslip angle (β) comparison

8.3. Yaw Rate Response

The yaw rate response was investigated to evaluate the rotational behavior of the vehicle during the drift maneuver.

The reverse Ackermann configuration exhibited lower variations in yaw rate compared with the other steering configurations. This indicates that the vehicle rotational motion was more consistent and experienced fewer sudden changes during the maneuver.

The results are consistent with the sideslip angle analysis, where the reverse Ackermann configuration showed a more stable lateral response.

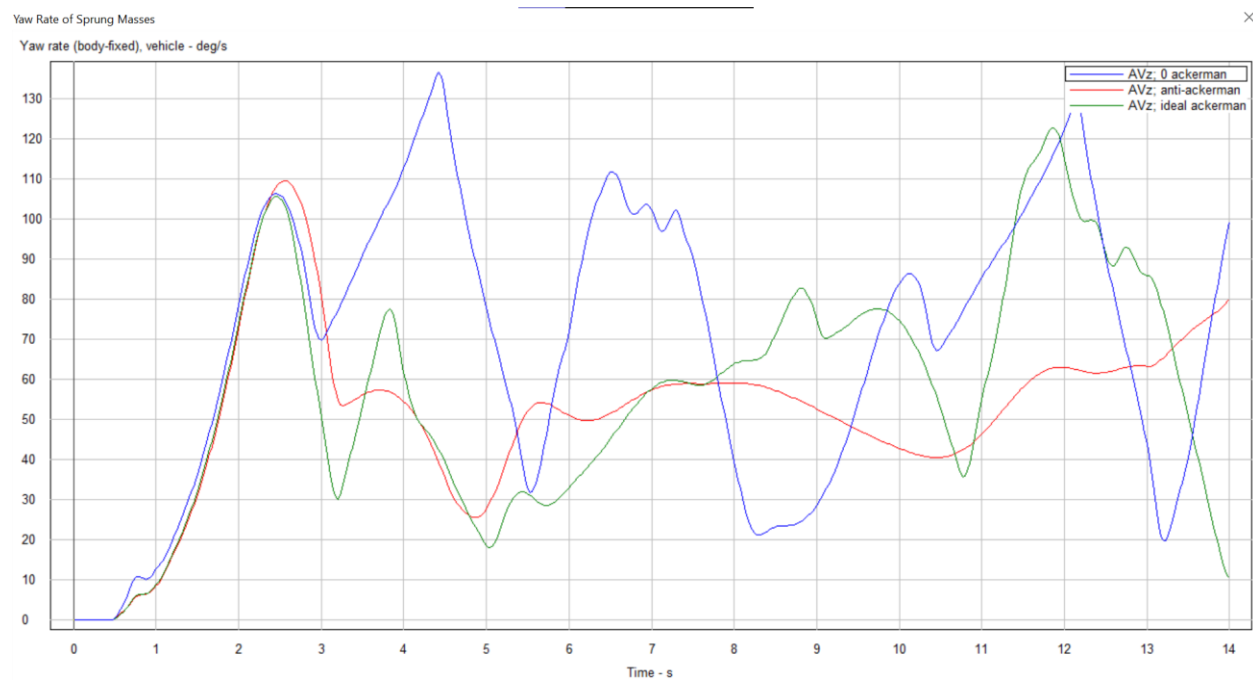


Figure 13 Yaw rate comparison for different steering geometries

8.4. Steering Torque Analysis

The steering torque required for each steering configuration was also evaluated.

The ideal Ackermann configuration generated higher steering torque compared with the other configurations. This behavior can be related to the different wheel steering angle relationships and the resulting tire forces during the maneuver.

The difference in steering torque indicates that modifying Ackermann characteristics can influence not only vehicle stability but also steering effort and driver feedback.

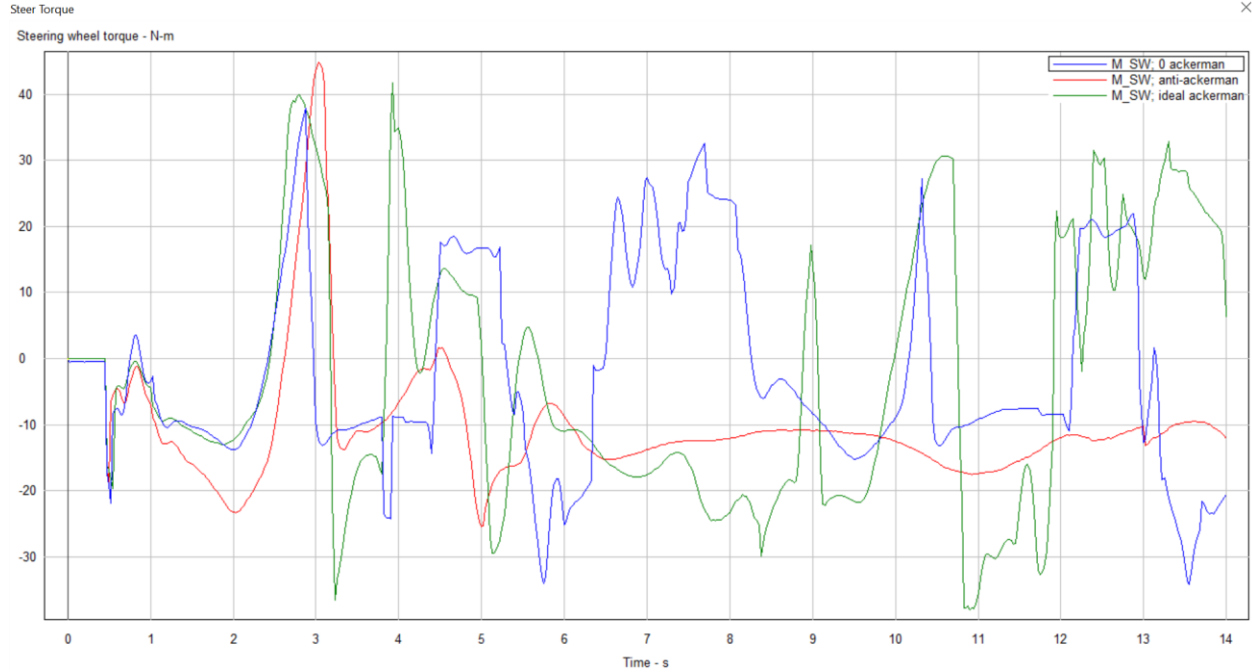


Figure 14 Steering torque comparison

8.5. Discussion of Simulation Results

The simulation results demonstrate that steering geometry has a significant influence on drift vehicle behavior.

Under the investigated conditions, the reverse Ackermann configuration provided the most stable response in terms of trajectory maintenance, sideslip angle variation, and yaw rate consistency. The parallel steering configuration showed less favorable behavior, while the conventional Ackermann configuration provided intermediate performance.

However, the obtained results are specific to the selected vehicle parameters, driving conditions, and track geometry. Different vehicle setups, tire characteristics, and drift scenarios may lead to different optimal steering characteristics.

The developed adjustable steering mechanism provides the capability to investigate and adapt steering geometry for different operating conditions, which can be beneficial for drift vehicle development

9. Prototype Assembly and Validation

9.1. Prototype Assembly

After completing the manufacturing process, the fabricated components were assembled with the original steering knuckle to evaluate the integration of the developed steering kit.

The assembly process was performed based on the CAD design developed from the reverse-engineered geometry. The compatibility between the modified components and the original suspension interfaces was verified during the assembly procedure.

The prototype assembly confirmed that the designed mechanism could be installed while maintaining the original mounting locations of the steering knuckle. This approach allowed the steering characteristics to be modified without requiring complete replacement of the original suspension system.

9.2. Mechanical Function Verification

Following the assembly process, the movement of the steering mechanism was inspected to verify its functionality.

The adjustable mechanism was evaluated to ensure that the tie-rod attachment point could be repositioned according to the designed configuration. The rotation of the adjustable insert changed the effective steering linkage geometry and allowed different steering characteristics to be generated.

The prototype inspection also focused on the available clearance between steering components during large steering angles. The final design provided sufficient clearance to achieve high-angle steering operation while avoiding mechanical interference between the steering components.

9.3. Prototype Evaluation

The manufactured prototype demonstrated the feasibility of the proposed adjustable steering concept.

The combination of reverse engineering, CAD-based design, and prototype manufacturing resulted in a functional steering modification system capable of altering steering geometry characteristics.

The prototype provided a practical platform for further investigation of steering configurations under different vehicle conditions.

Although the present validation focused on mechanical integration and functionality, additional experimental testing under actual driving conditions can provide further insight into the relationship between steering geometry and vehicle dynamic behavior.

10. Conclusion

In this project, a modular high-angle steering kit was designed and developed for a Nissan 350Z drift application. The main objective was to increase steering angle capability and provide the possibility of modifying Ackermann steering characteristics through an adjustable mechanical mechanism.

The development process started with three-dimensional scanning of the original steering knuckle, followed by reverse engineering using Geomagic and CAD reconstruction in SolidWorks. Based on the reconstructed geometry, an adjustable steering mechanism was designed to modify the tie-rod attachment location relative to the steering axis.

The proposed mechanism utilized a rotating insert with an obround adjustment slot, allowing modification of the steering linkage geometry. The final prototype was manufactured using CNC machining, laser cutting, and CNC bending processes. The developed system achieved a maximum steering angle of approximately 41 degrees while maintaining compatibility with the original steering components.

The influence of different steering geometries was investigated using CarSim simulations under identical drift-oriented conditions. The simulation results showed that the reverse Ackermann configuration provided more stable behavior compared with the other investigated configurations, including lower variations in vehicle sideslip angle and yaw rate, as well as improved trajectory maintenance.

The results demonstrate that adjustable steering geometry can be an effective approach for adapting vehicle steering characteristics to drift applications. The developed modular steering system provides a flexible platform for further optimization and investigation of steering configurations under different driving conditions.

Future Work

Although the developed system successfully demonstrated the capability of modifying steering geometry, several improvements can be considered for future development.

Future work may include:

- Experimental testing under real driving conditions.
- Structural analysis of the manufactured components using finite element analysis (FEA).
- Optimization of steering geometry for different tracks and driving styles.
- Investigation of the effect of tire characteristics and suspension setup on steering geometry selection.
- Development of interchangeable inserts for track-specific drift configurations